THE TRIANGULATED CATEGORY OF PERFECT COMPLEXES OVER A FILTERED COLIMIT OF RINGS IS THE COLIMIT OF THE CATEGORIES OVER THE STAGES

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The chapter file pinned-original-source.tex carries only \input{preamble} on its first line, so no preamble text lives inside the chapter. The retained file sources/stacks-more-algebra/retained-preamble.tex is a mechanical, line-for-line copy of the project's single shared preamble as staged in the snapshot; nothing was added, removed or redefined and no package was added.

Editorial note. The source URL is the source-level Stacks Project URL, as required when more than one card is produced from one source; the per-claim tag is carried by claim_key.

Editorial note. The six references inside the counted proof span are recorded as non-load-bearing navigation pointers.

Allowed standard prerequisites (editorial).

- Filtered colimits of rings and modules, and their exactness. used throughout the counted proof span. Is what allows finite data over the colimit to be descended to a stage.
- Perfect complexes as bounded complexes of finite projective modules. used in the counted proof span. Fixes the form in which the descending object is presented.

Source reference. Original lemma 05N7: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 7296–7296. Cross-chapter navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 00DB: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 491–491. Cross-chapter navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 00NX: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 875–875. Cross-chapter navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 05LI: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 7237–7237. Cross-chapter navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Date: stacks-project:more-algebra.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 064B: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 7860–7860. Cross-chapter navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 14380: sources/stacks-more-algebra/pinned-original-source.tex, source lines 14380–14380. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Lemma 0.2. *Let $R = \operatorname{colim}_{i \in I} R_i$ be a filtered colimit of rings.*

(1) *Given a perfect K in $D(R)$ there exists an $i \in I$ and a perfect K_i in $D(R_i)$ such that $K \cong K_i \otimes^{\mathbf{L}}_{R_i} R$ in $D(R)$.*
(2) *Given $0 \in I$ and $K_0, L_0 \in D(R_0)$ with K_0 perfect, we have*

$$\operatorname{Hom}_{D(R)}(K_0 \otimes^{\mathbf{L}}_{R_0} R, L_0 \otimes^{\mathbf{L}}_{R_0} R) = \operatorname{colim}_{i \geq 0} \operatorname{Hom}_{D(R_i)}(K_0 \otimes^{\mathbf{L}}_{R_0} R_i, L_0 \otimes^{\mathbf{L}}_{R_0} R_i)$$

In other words, the triangulated category of perfect complexes over R is the colimit of the triangulated categories of perfect complexes over R_i.

Proof. We will use the results of Algebra, Lemmas 05LI and 05N7 without further mention. These lemmas in particular say that the category of finitely presented R-modules is the colimit of the categories of finitely presented R_i-modules. Since finite projective modules can be characterized as summands of finite free modules (Algebra, Lemma 00NX) we see that the same is true for the category of finite projective modules. This proves (1) by our definition of perfect objects of $D(R)$.

To prove (2) we may represent K_0 by a bounded complex $K_0^\bullet$ of finite projective R_0-modules. We may represent L_0 by a K-flat complex $L_0^\bullet$ (Lemma 14380). Then we have

$$\operatorname{Hom}_{D(R)}(K_0 \otimes^{\mathbf{L}}_{R_0} R, L_0 \otimes^{\mathbf{L}}_{R_0} R) = \operatorname{Hom}_{K(R)}(K_0^\bullet \otimes_{R_0} R, L_0^\bullet \otimes_{R_0} R)$$

by Derived Categories, Lemma 064B. Similarly for the Hom with R replaced by R_i. Since in the right hand side only a finite number of terms are involved, since

$$\operatorname{Hom}_R(K_0^p \otimes_{R_0} R, L_0^q \otimes_{R_0} R) = \operatorname{colim}_{i \geq 0} \operatorname{Hom}_{R_i}(K_0^p \otimes_{R_0} R_i, L_0^q \otimes_{R_0} R_i)$$

by the lemmas cited at the beginning of the proof, and since filtered colimits are exact (Algebra, Lemma 00DB) we conclude that (2) holds as well. □